

(12) PATENT APPLICATION PUBLICATION

(21) Application No.202541088551 A

(19) INDIA

(22) Date of filing of Application :17/09/2025

(43) Publication Date : 17/10/2025

(54) Title of the invention : A MULTIMODAL SYSTEM FOR GESTURE RECOGNITION

(51) International classification	:G06F0003010000, G06F0003048860, G06N0020000000, G06V0040200000, G06F0003160000	(71) Name of Applicant : 1)NEW HORIZON COLLEGE OF ENGINEERING Address of Applicant :New Horizon College of Engineering, New Horizon Knowledge Park, Marathalli, Outer ring road, Bengaluru-560103. Bangalore Karnataka India
(31) Priority Document No	:NA	(72) Name of Inventor :
(32) Priority Date	:NA	1)Dr.Jimsha K Mathew
(33) Name of priority country	:NA	2)Yashas D
(86) International Application No	:	3)Pratibha Prakash Machakanur
Filing Date	:01/01/1900	4)M Shivani Kashyap
(87) International Publication No	: NA	5)Vinay Prasad K
(61) Patent of Addition to Application Number	:NA	
Filing Date	:NA	
(62) Divisional to Application Number	:NA	
Filing Date	:NA	

(57) Abstract :

The system Multimodal System for Gesture Recognition enables contactless human-computer interaction through integrated modules: Virtual Mouse with Voice Assistance, Virtual Calculator, Virtual Canvas, and Virtual Keyboard. It uses OpenCV and Mediapipe for real-time video processing and 21-point hand landmark detection, allowing precise gesture control. Voice commands are processed using Natural Language Processing (NLP), enabling intuitive system interaction. A hybrid rule-based and machine learning engine dynamically adapts to user behavior. The calculator interprets hand-drawn equations using Generative AI, while the canvas allows gesture-based drawing with shape and color selection. A custom-trained model with over 15,000 gesture samples, enhanced using temporal modeling, gesture morphing, and background blending, ensures high accuracy. Supporting both marker-free and marker-based input, the system enhances accessibility, making it ideal for education, creativity, assistive technology, and hands-free computing.

No. of Pages : 18 No. of Claims : 6